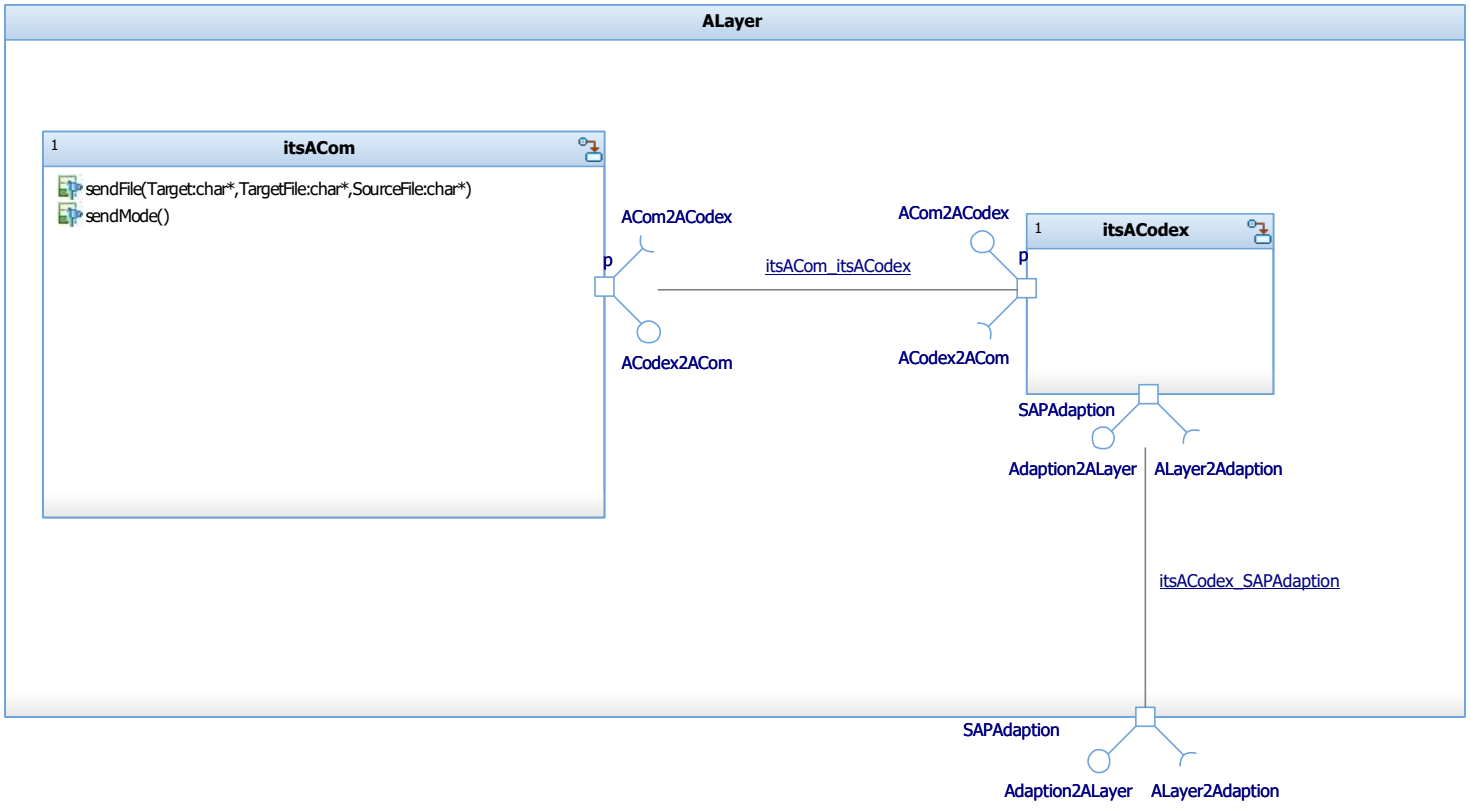
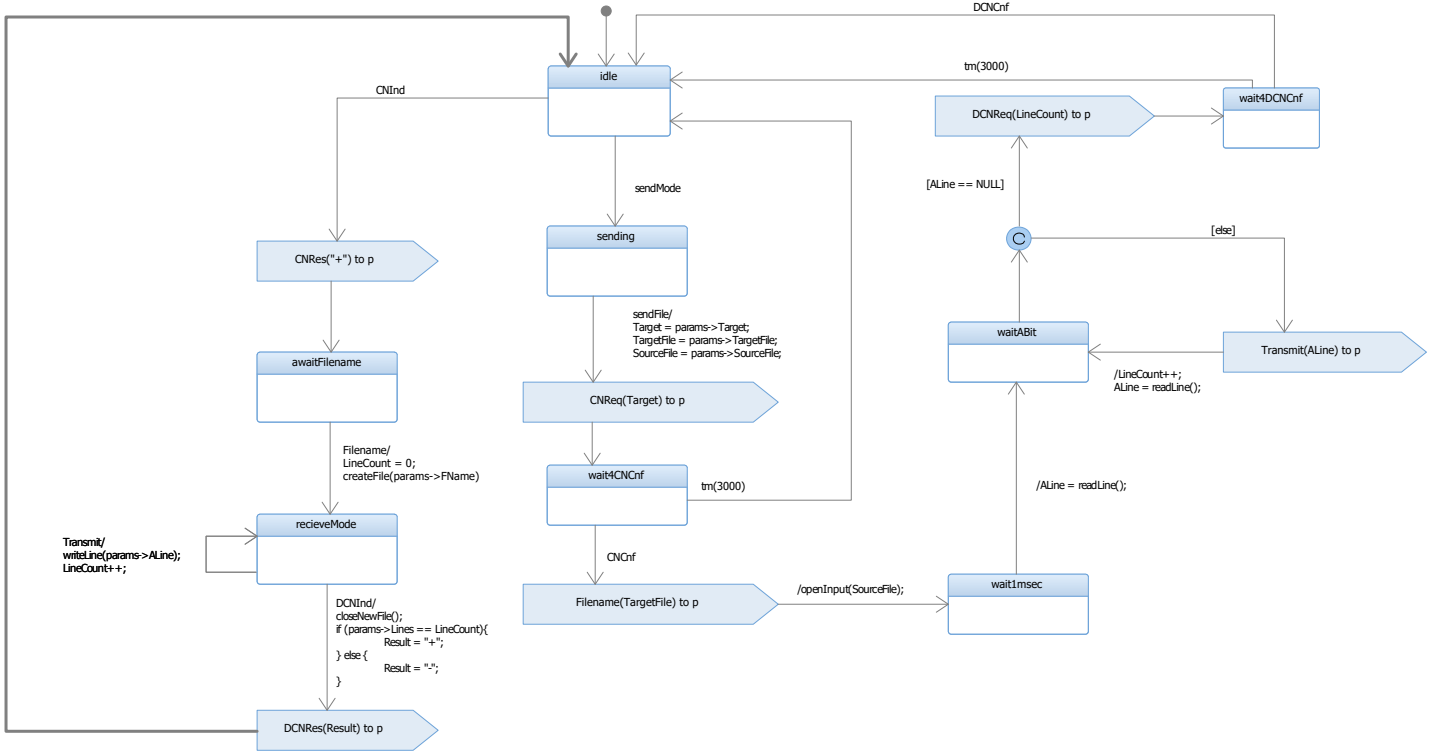
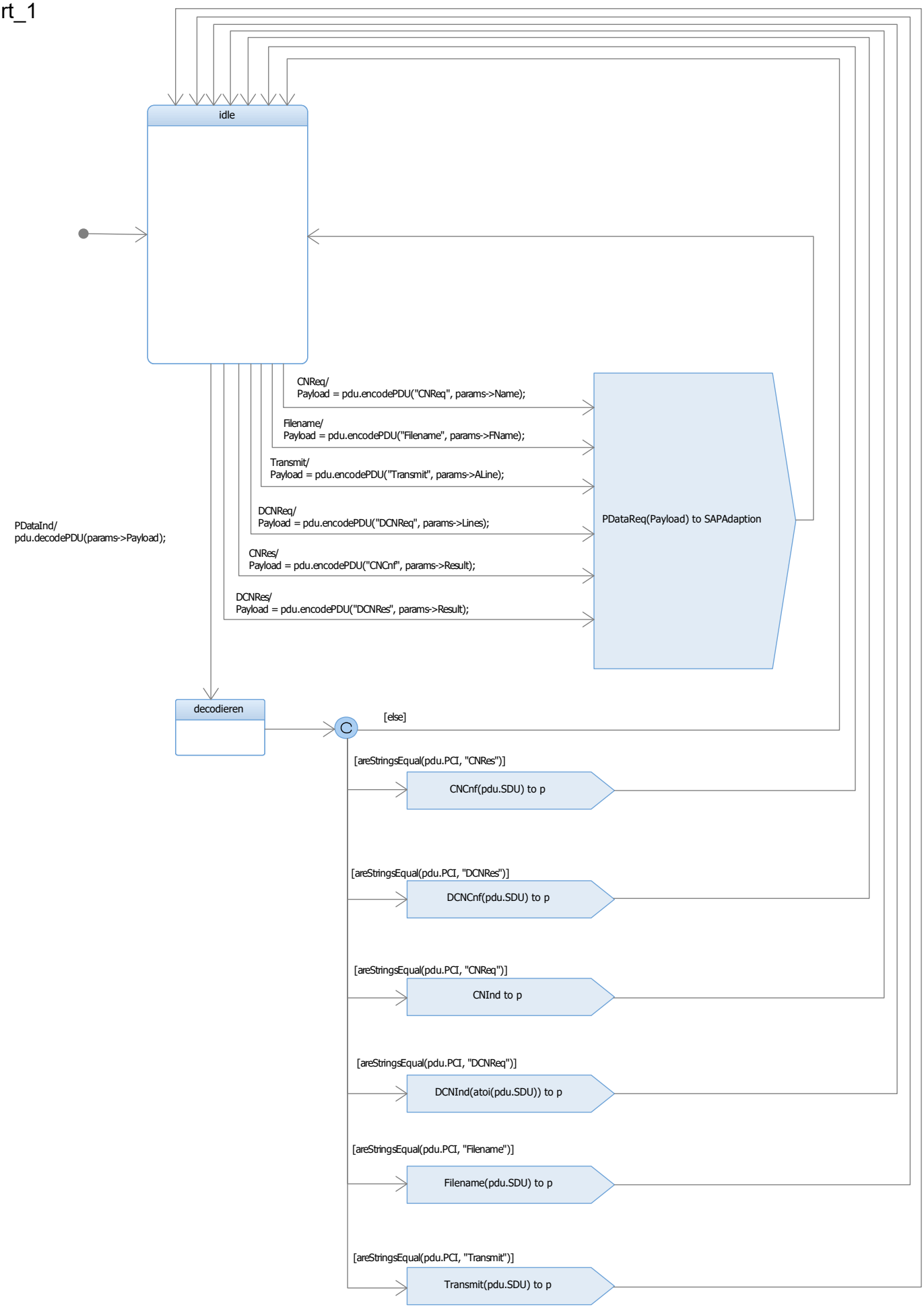
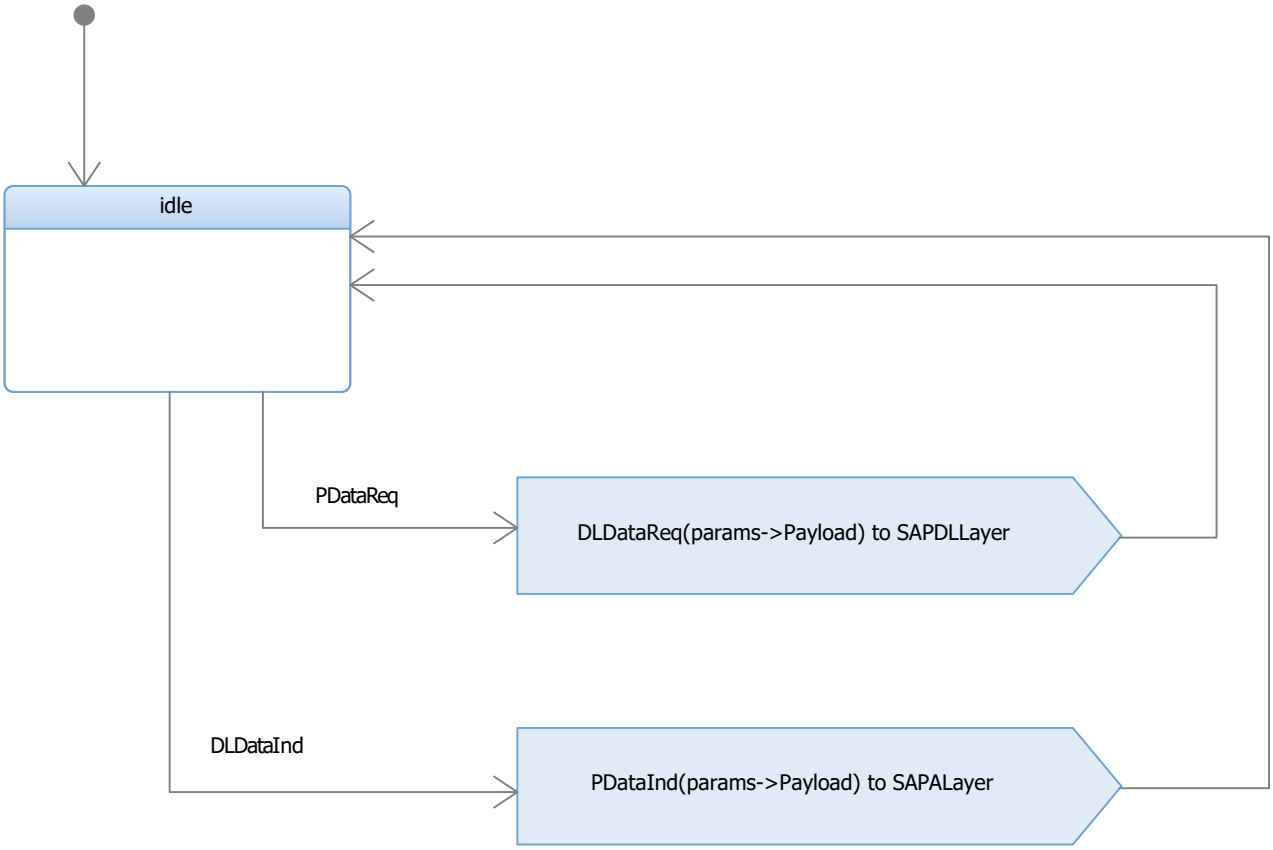
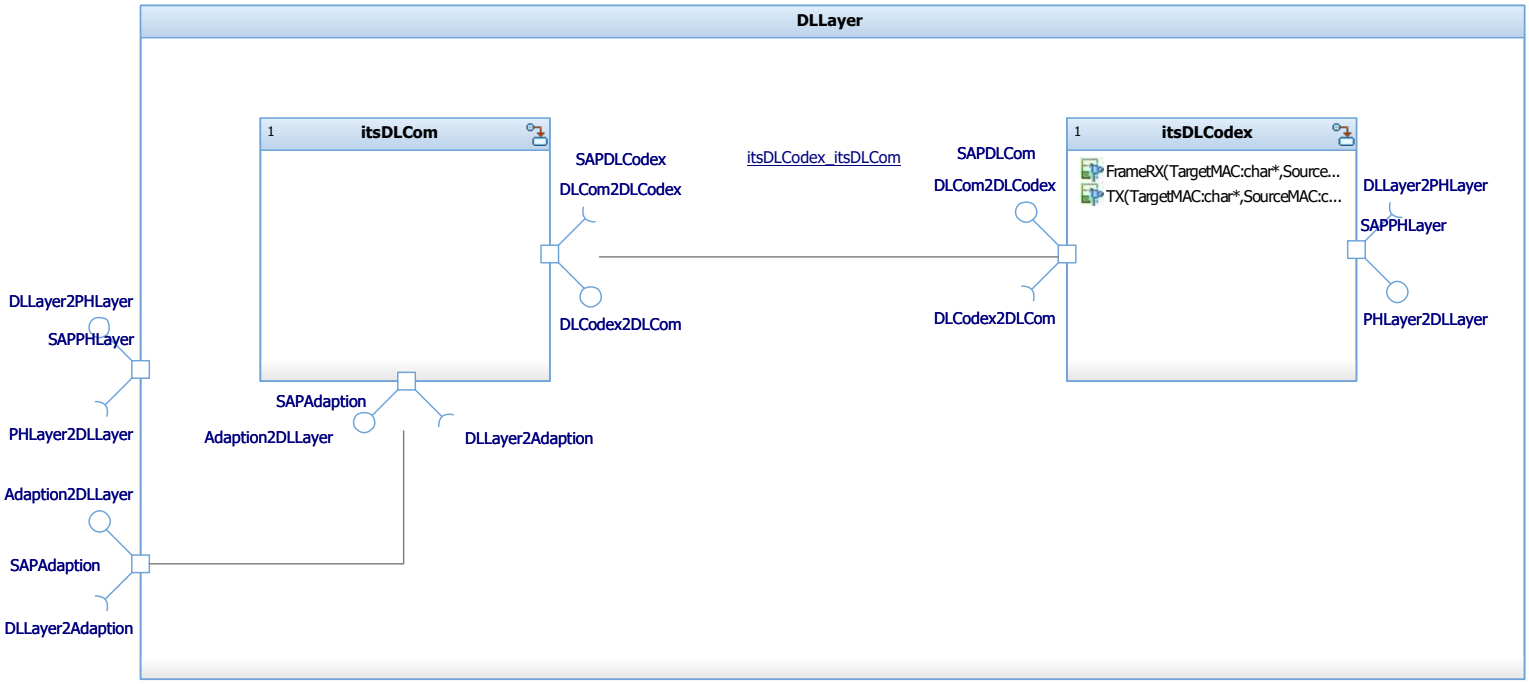

ALayer
1
itsACom
sendFile(Target:char*,TargetFile:char*,SourceFile:char*)
sendMode()
ACom2ACodex
p
ACodex2ACom
itsACom_itsACodex
ACom2ACodex
p
ACodex2ACom
1
itsACodex
SAPAdaption
Adaption2ALayer
ALayer2Adaption
itsACodex_SAPAdaption
SAPAdaption
Adaption2ALayer
ALayer2Adaption

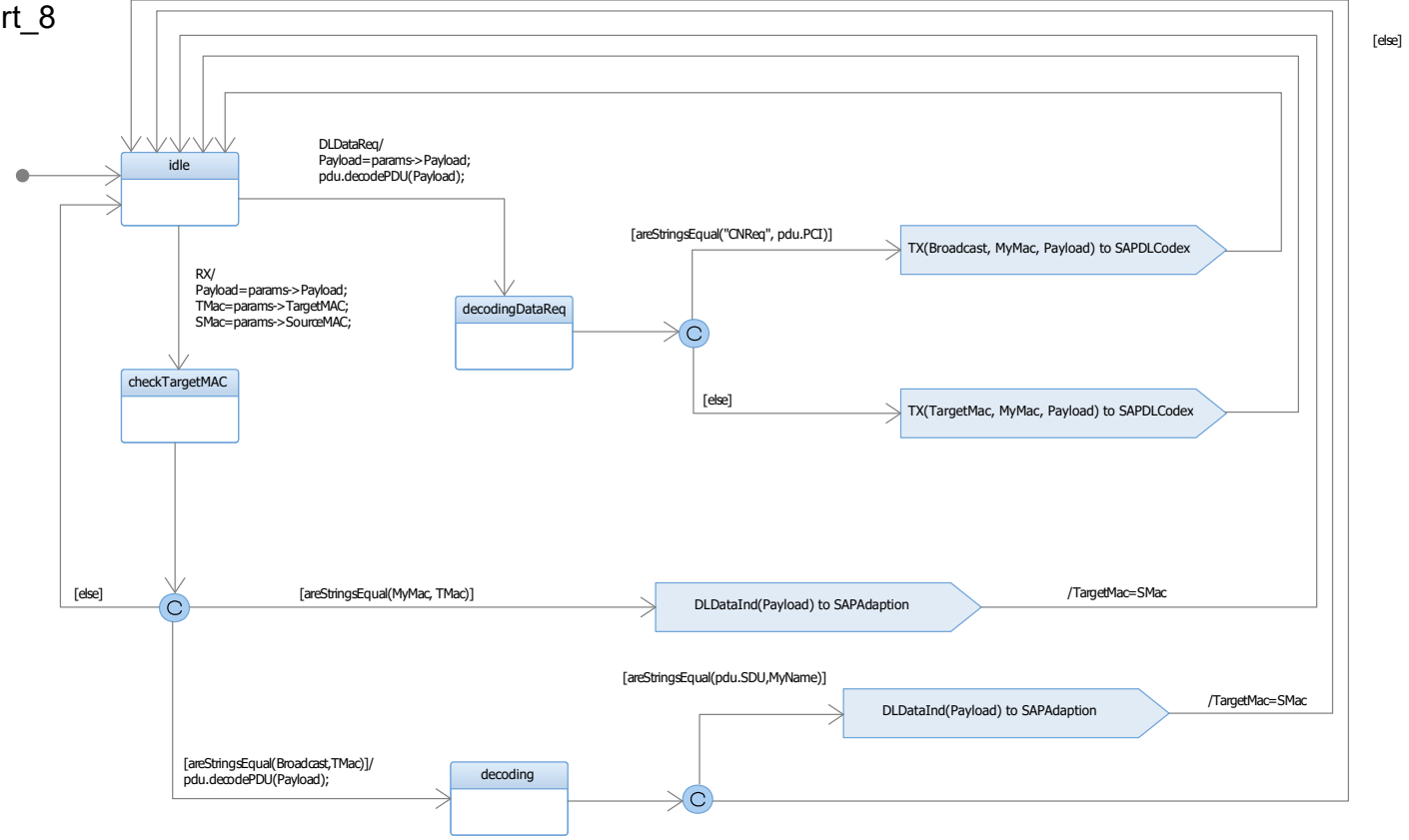

DCNCnf
idle
tm(3000)
wait4DCNCnf
CNInd
DCNReq(LineCount) to p
sendMode
[ALine == NULL]
sending
[else]
CNRes("+") to p
sendFile/
Target = params->Target;
TargetFile = params->TargetFile;
SourceFile = params->SourceFile;
waitABit
Transmit(ALine) to p
awaitFilename
/LineCount++;
ALine = readLine();
CNReq(Target) to p
Filename/
LineCount = 0;
createFile(params->FName)
wait4CNCnf
tm(3000)
/ALine = readLine();
recieveMode
Transmit/
writeLine(params->ALine);
LineCount++;
CNCnf
Filename(TargetFile) to p
/openInput(SourceFile);
wait1msec
DCNInd/
closeNewFile();
if (params->Lines == LineCount){
Result = "+";
} else {
Result = "-";
}
DCNRes(Result) to p


idle
CNReq/
Payload = pdu.encodePDU("CNReq", params->Name);
Filename/
Payload = pdu.encodePDU("Filename", params->FName);
Transmit/
Payload = pdu.encodePDU("Transmit", params->ALine);
DCNReq/
Payload = pdu.encodePDU("DCNReq", params->Lines);
CNRes/
Payload = pdu.encodePDU("CNCnf", params->Result);
DCNRes/
Payload = pdu.encodePDU("DCNRes", params->Result);
PDataReq(Payload) to SAPAdaption
PDataInd/
pdu.decodePDU(params->Payload);
decodieren
[else]
[areStringsEqual(pdu.PCI, "CNRes")]
CNCnf(pdu.SDU) to p
[areStringsEqual(pdu.PCI, "DCNRes")]
DCNCnf(pdu.SDU) to p
[areStringsEqual(pdu.PCI, "CNReq")]
CNInd to p
[areStringsEqual(pdu.PCI, "DCNReq")]
DCNInd(atoi(pdu.SDU)) to p
[areStringsEqual(pdu.PCI, "Filename")]
Filename(pdu.SDU) to p
[areStringsEqual(pdu.PCI, "Transmit")]
Transmit(pdu.SDU) to p

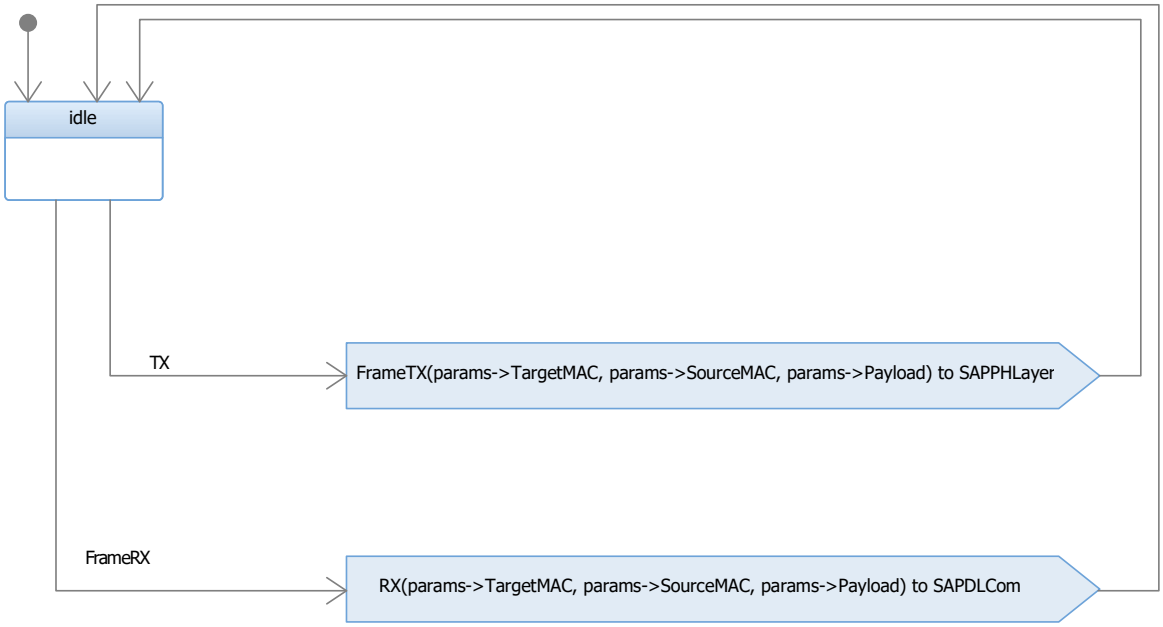

idle
PDataReq
DLDataReq(params->Payload) to SAPDLLayer
DLDataInd
PDataInd(params->Payload) to SAPALayer

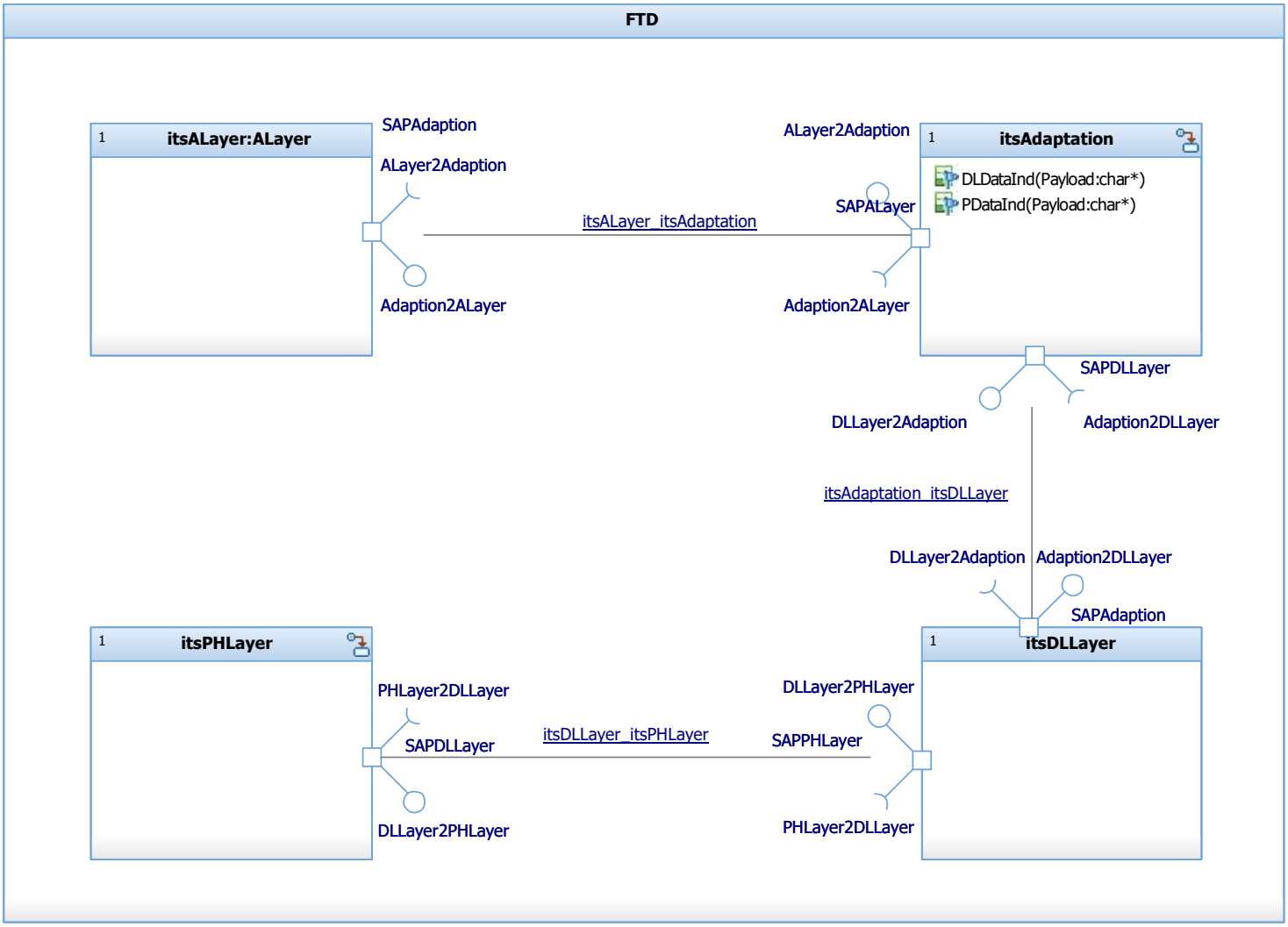

DLLayer
1
itsDLCom
SAPDLCodex
DLCom2DLCodex
DLCodex2DLCom
itsDLCodex_itsDLCom
SAPDLCom
DLCom2DLCodex
DLCodex2DLCom
1
itsDLCodex
FrameRX(TargetMAC:char*,Source...
TX(TargetMAC:char*,SourceMAC:c...
DLLayer2PHLayer
SAPPHLayer
PHLayer2DLLayer
DLLayer2PHLayer
SAPPHLayer
PHLayer2DLLayer
SAPAdaption
Adaption2DLLayer
DLLayer2Adaption
Adaption2DLLayer
SAPAdaption
DLLayer2Adaption

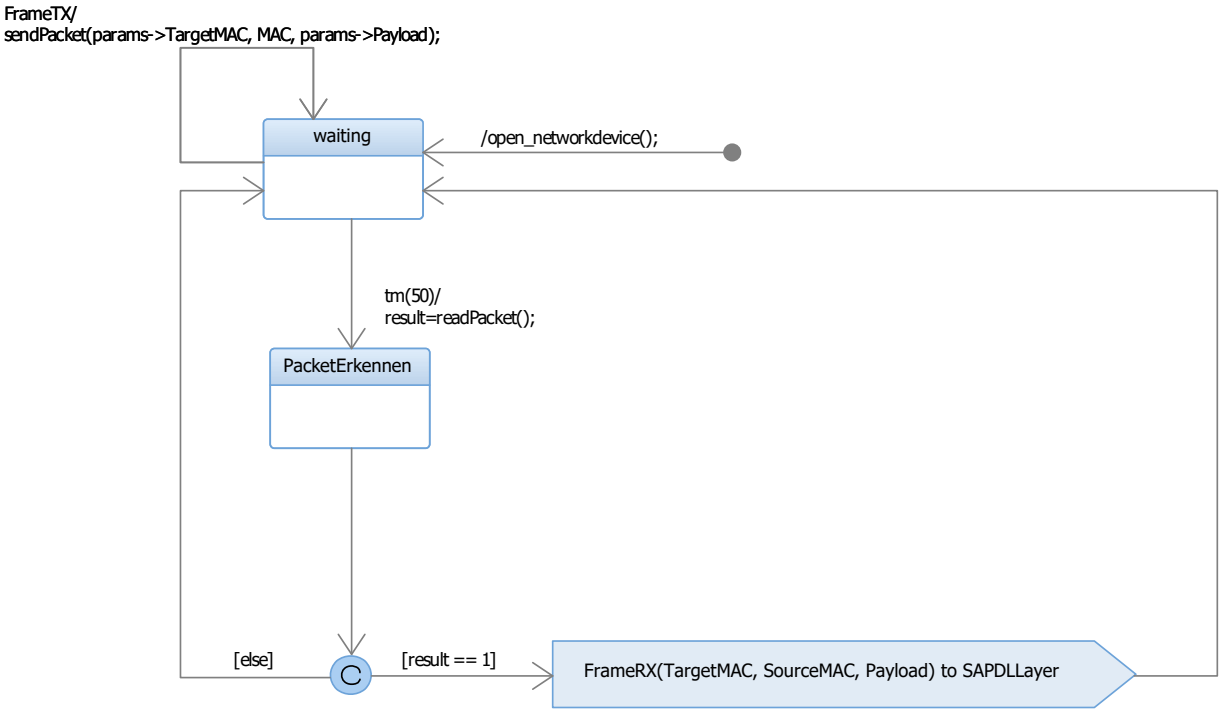
statechart_8

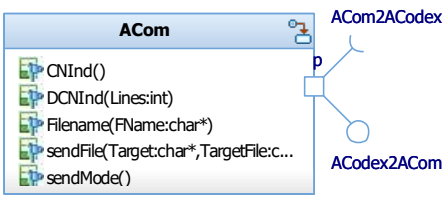


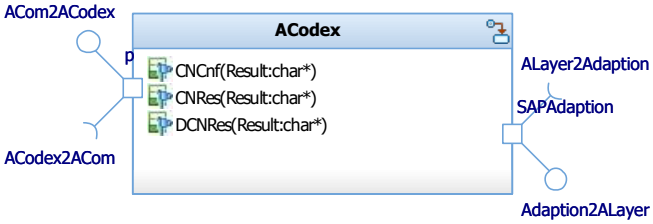
statechart_1

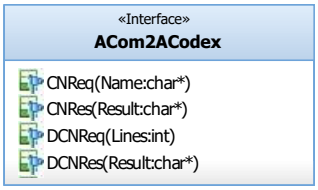


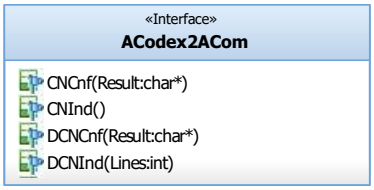

FTD
1
itsALayer:ALayer
SAPAdaption
ALayer2Adaption
Adaption2ALayer
itsALayer_itsAdaptation
ALayer2Adaption
SAPALayer
Adaption2ALayer
1
itsAdaptation
DLDataInd(Payload:char*)
PDataInd(Payload:char*)
SAPDLLayer
DLLayer2Adaption
Adaption2DLLayer
itsAdaptation_itsDLLayer
DLLayer2Adaption
Adaption2DLLayer
SAPAdaption
1
itsPHLayer
PHLayer2DLLayer
SAPDLLayer
DLLayer2PHLayer
itsDLLayer_itsPHLayer
DLLayer2PHLayer
SAPPHLayer
PHLayer2DLLayer
1
itsDLLayer


FrameTX/
sendPacket(params->TargetMAC, MAC, params->Payload);
waiting
/open_networkdevice();
tm(50)/
result=readPacket();
PacketErkennen
[else]
C
[result == 1]
FrameRX(TargetMAC, SourceMAC, Payload) to SAPDLLayer

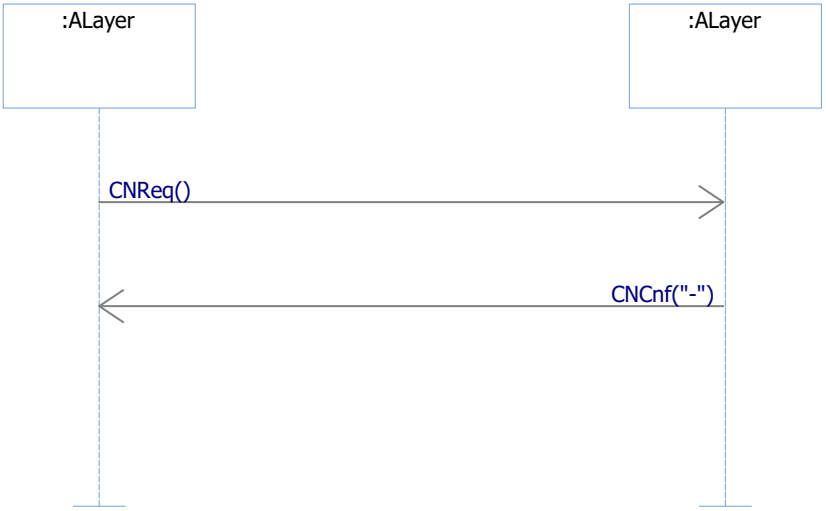

ACom
CNInd()
DCNInd(Lines:int)
Filename(FName:char*)
sendFile(Target:char*,TargetFile:c...
sendMode()
p
ACom2ACodex
ACodex2ACom


ACom2ACodex
p
ACodex2ACom
ACodex
CNCnf(Result:char*)
CNRes(Result:char*)
DCNRes(Result:char*)
ALayer2Adaption
SAPAdaption
Adaption2ALayer

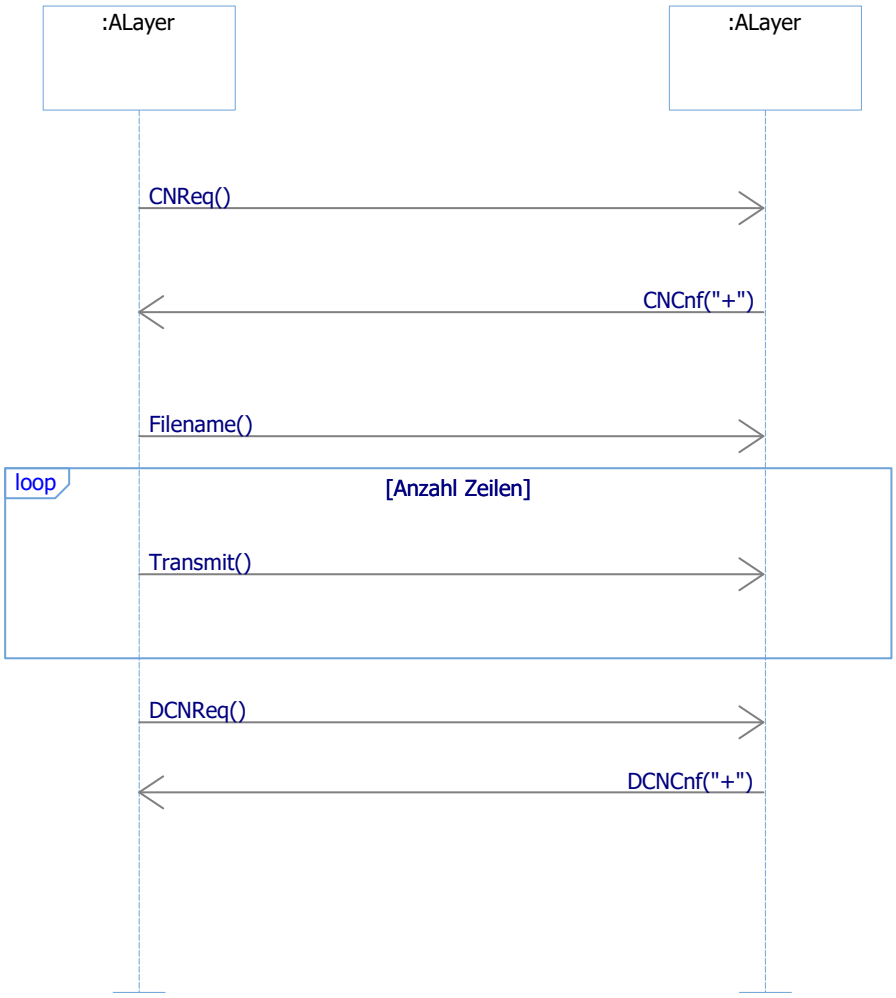

«Interface»
ACom2ACodex
CNReq(Name:char*)
CNRes(Result:char*)
DCNReq(Lines:int)
DCNRes(Result:char*)


«Interface»
ACodex2ACom
CNCnf(Result:char*)
CNInd()
DCNCnf(Result:char*)
DCNInd(Lines:int)

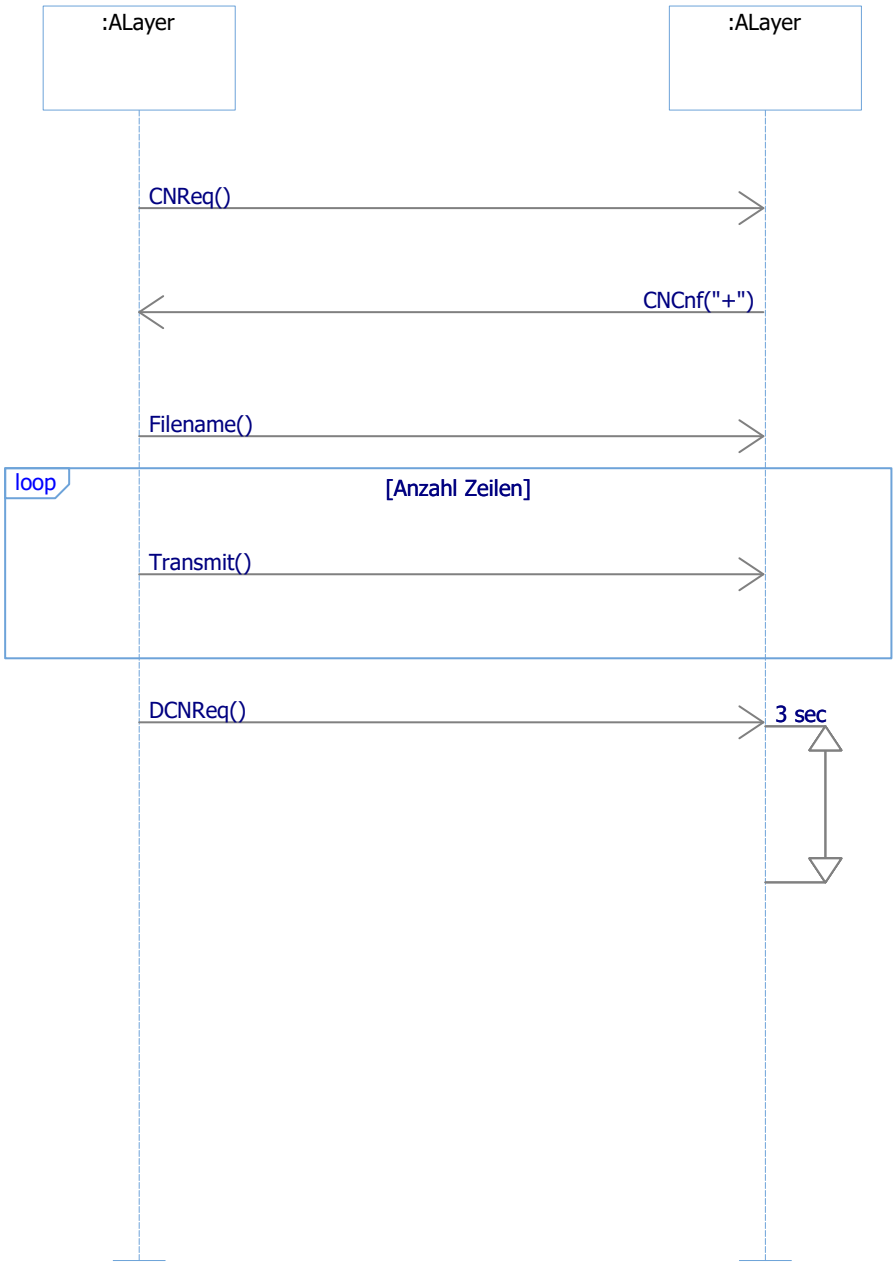
CNCnf schlaegt 1mal fehl



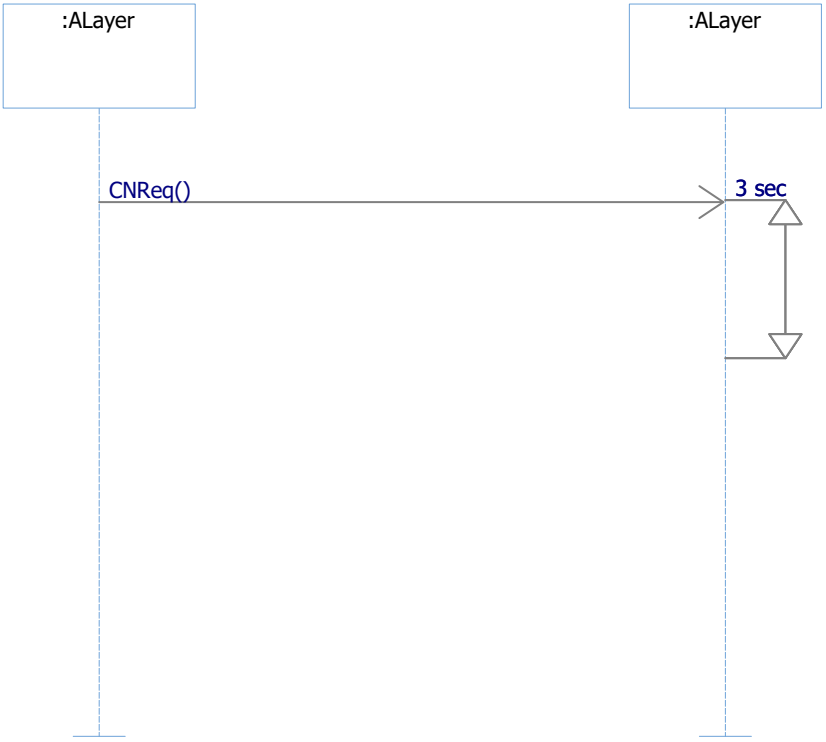
Erfolgreicher Ablauf



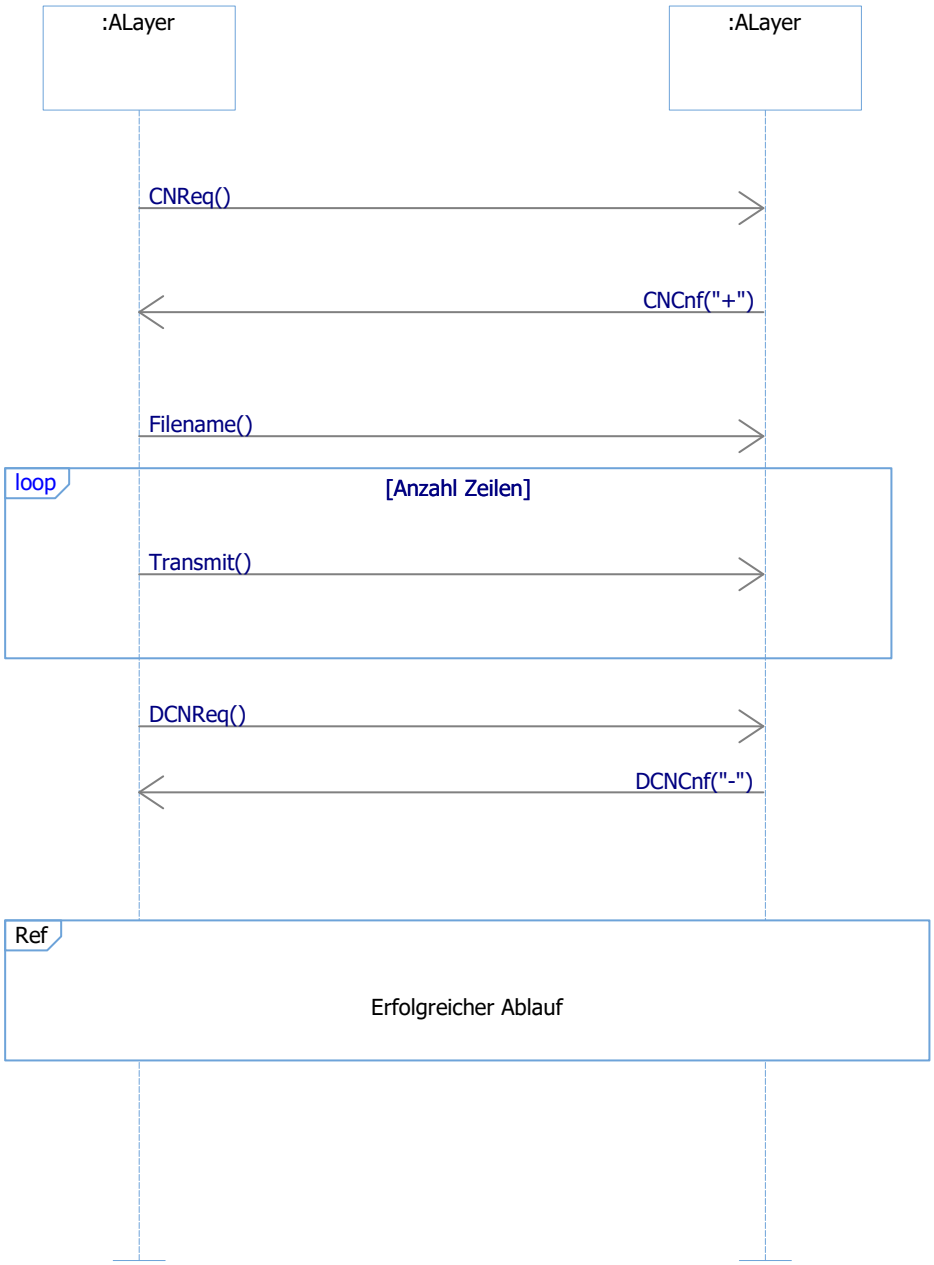
DCNCnf schlaegt fehl



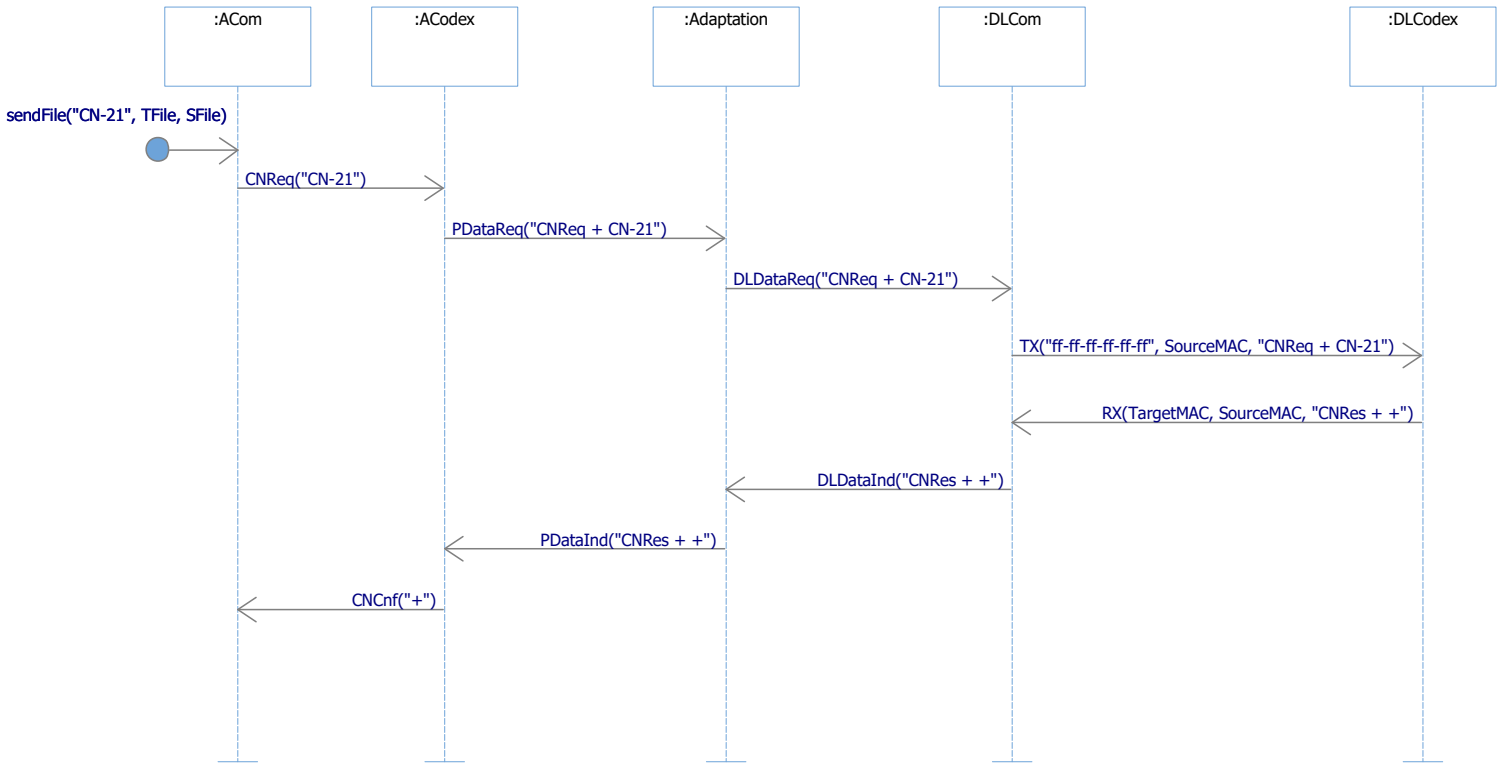
CNCnf schlaegt timeout

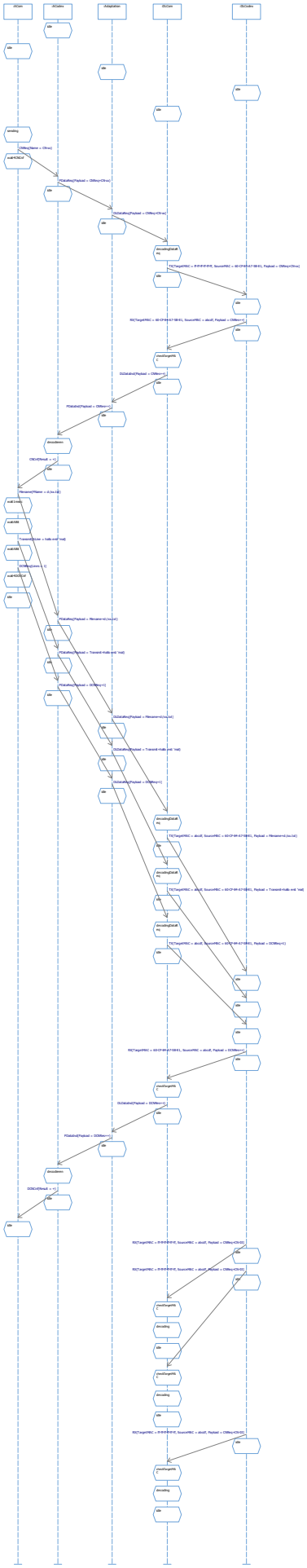


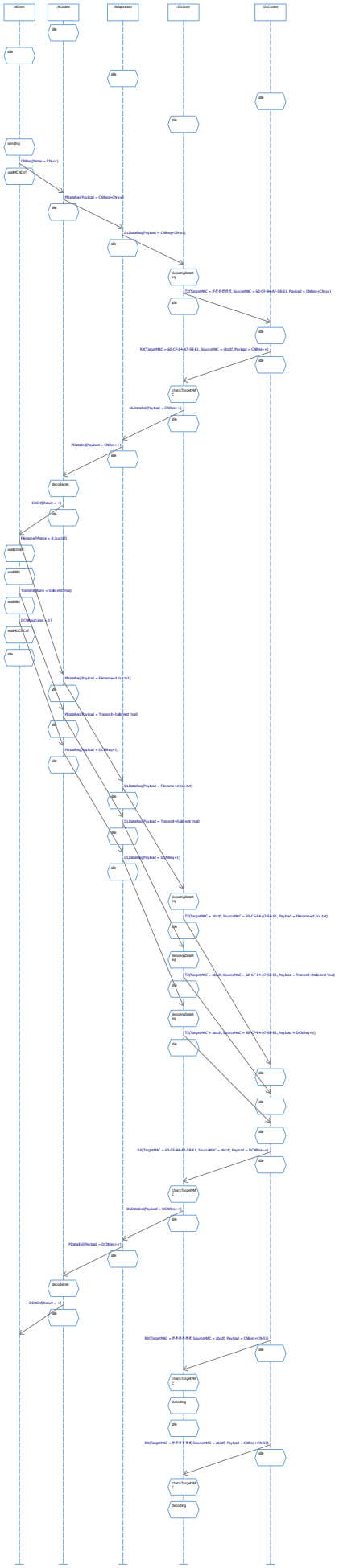
DCNCnf schlaegt timeout



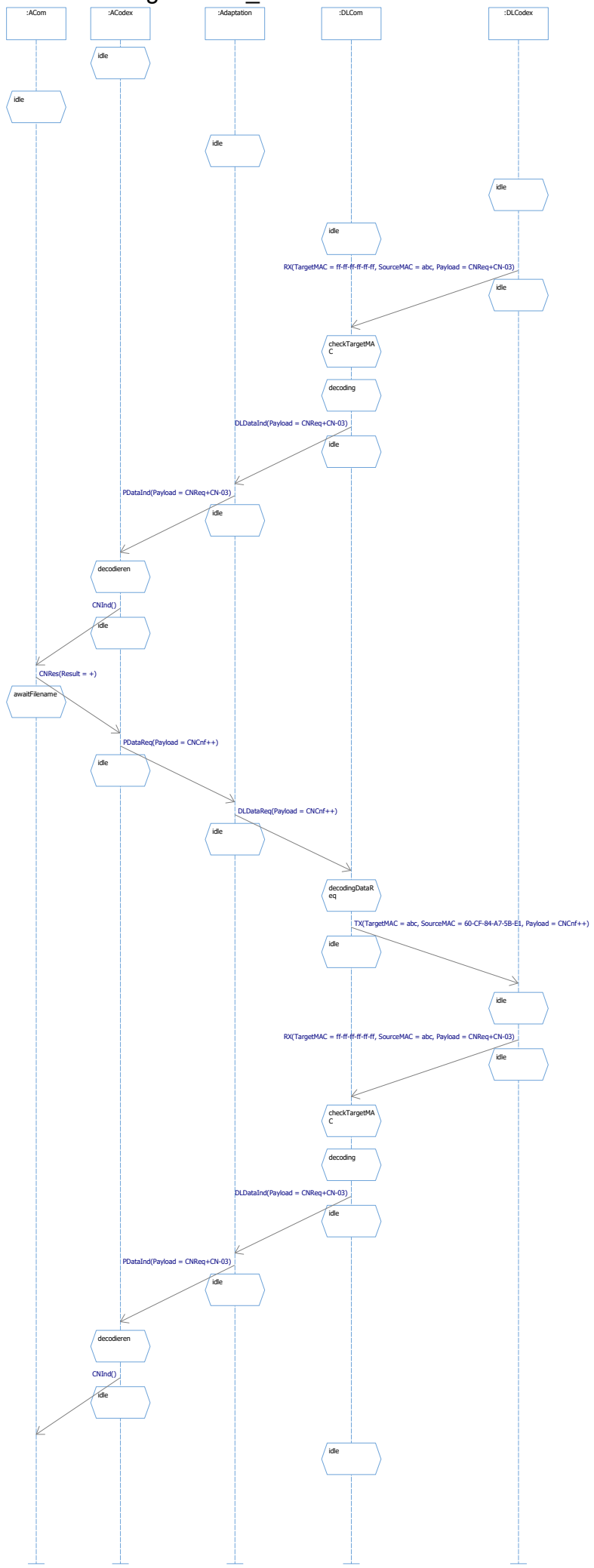
Verbindungsaufbau

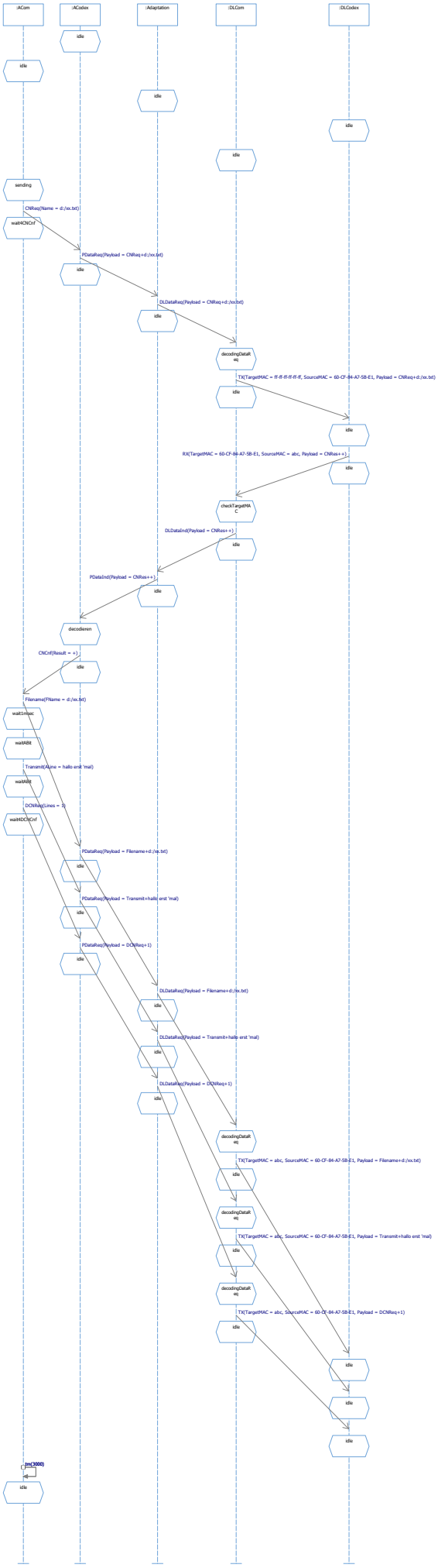



JCCom
JCCodec
Adaptation
DLCom
DLCodec
idle
sending
CMReq(Name = CHrxc)
wait4CNCnf
FDataReq(Payload = CMReq=CHrxc)
idle
DLDataReq(Payload = CMReq=CHrxc)
idle
decoding(Data Rx)
TX(Target/MAC = ff-ff-ff-ff-ff-ff, Source/MAC = 60-CF-8F-A7-5B-E1, Payload = CMReq=CHrxc)
idle
idle
RX(Target/MAC = 60-CF-8F-A7-5B-E1, Source/MAC = abcdf, Payload = CMResp=1)
idle
checkTargetMAC
DLDataInd(Payload = CMResp=1)
idle
FDataInd(Payload = CMResp=1)
idle
decoderesp
CNCnf(Result = 1)
idle
Filename(FName = d.txt,16)
wait1MB
wait4MB
Transmit(Size = hallo end "end)
wait1MB
DCMReq(Lines = 1)
wait4DCCnf
idle
FDataReq(Payload = Filename=d.txt,16)
idle
FDataReq(Payload = Transmit=hallo end "end)
idle
FDataReq(Payload = DCMReq=1)
idle
DLDataReq(Payload = Filename=d.txt,16)
idle
DLDataReq(Payload = Transmit=hallo end "end)
idle
DLDataReq(Payload = DCMReq=1)
idle
decoding(Data Rx)
TX(Target/MAC = abcdf, Source/MAC = 60-CF-8F-A7-5B-E1, Payload = Filename=d.txt,16)
idle
decoding(Data Rx)
TX(Target/MAC = abcdf, Source/MAC = 60-CF-8F-A7-5B-E1, Payload = Transmit=hallo end "end)
idle
decoding(Data Rx)
TX(Target/MAC = abcdf, Source/MAC = 60-CF-8F-A7-5B-E1, Payload = DCMReq=1)
idle
idle
idle
idle
RX(Target/MAC = 60-CF-8F-A7-5B-E1, Source/MAC = abcdf, Payload = DCMResp=1)
idle
checkTargetMAC
DLDataInd(Payload = DCMResp=1)
idle
FDataInd(Payload = DCMResp=1)
idle
decoderesp
DCMCnf(Result = 1)
idle
idle
RX(Target/MAC = ff-ff-ff-ff-ff-ff, Source/MAC = abcdf, Payload = CMReq=CHrxc)
idle
RX(Target/MAC = ff-ff-ff-ff-ff-ff, Source/MAC = abcdf, Payload = CMReq=CHrxc)
idle
checkTargetMAC
decoding
idle
checkTargetMAC
decoding
idle
RX(Target/MAC = ff-ff-ff-ff-ff-ff, Source/MAC = abcdf, Payload = CMReq=CHrxc)
idle
checkTargetMAC
decoding
idle


JCCom
JCCodec
JAdaptation
DLCom
DLCodec
idle
idle
idle
idle
idle
sending
CMReq(Name = Ch-xx)
waitHCNCnf
PDataReq(Payload = CMReq+Ch-xx)
idle
DLDataReq(Payload = CMReq+Ch-xx)
idle
decodingdatafr
ame
TX(TargetMAC = ff-ff-ff-ff-ff-ff, SourceMAC = 60-CF-84-A7-5B-E1, Payload = CMReq+Ch-xx)
idle
idle
RX(TargetMAC = 60-CF-84-A7-5B-E1, SourceMAC = abcdf, Payload = CMAck+1)
idle
checkTargetMA
C
DLDataInd(Payload = CMAck+1)
idle
PDataInd(Payload = CMAck+1)
idle
decoderen
CNCnf(Result = 1)
idle
Filename(Name = d.txx.txt)
waitConnect
waitAck
Transmit(Name = talk-ent'mal)
waitAck
DCMReq(Axx = 1)
waitHCNCnf
idle
PDataReq(Payload = Filename=d.txx.txt)
idle
PDataReq(Payload = Transmit+talk-ent'mal)
idle
PDataReq(Payload = DCMReq+1)
idle
DLDataReq(Payload = Filename=d.txx.txt)
idle
DLDataReq(Payload = Transmit+talk-ent'mal)
idle
DLDataReq(Payload = DCMReq+1)
idle
decodingdatafr
ame
TX(TargetMAC = abcdf, SourceMAC = 60-CF-84-A7-5B-E1, Payload = Filename=d.txx.txt)
idle
decodingdatafr
ame
TX(TargetMAC = abcdf, SourceMAC = 60-CF-84-A7-5B-E1, Payload = Transmit+talk-ent'mal)
idle
decodingdatafr
ame
TX(TargetMAC = abcdf, SourceMAC = 60-CF-84-A7-5B-E1, Payload = DCMReq+1)
idle
idle
idle
idle
RX(TargetMAC = 60-CF-84-A7-5B-E1, SourceMAC = abcdf, Payload = DCMAck+1)
idle
checkTargetMA
C
DLDataInd(Payload = DCMAck+1)
idle
PDataInd(Payload = DCMAck+1)
idle
decoderen
DCNCnf(Result = 1)
idle
RX(TargetMAC = ff-ff-ff-ff-ff-ff, SourceMAC = abcdf, Payload = CMReq+Ch-01)
idle
checkTargetMA
C
decoding
idle
RX(TargetMAC = ff-ff-ff-ff-ff-ff, SourceMAC = abcdf, Payload = CMReq+Ch-02)
idle
checkTargetMA
C
decoding

Animated Verbindungsaufbau_2




:ACom
:ACodex
:Adaptation
:DLCom
:DLCodex
idle
sending
CNReq(Name = d:/xx.txt)
wait4CNCnf
PDataReq(Payload = CNReq+d:/xx.txt)
DLDataReq(Payload = CNReq+d:/xx.txt)
decodingDataReq
TX(TargetMAC = ff-ff-ff-ff-ff-ff, SourceMAC = 60-CF-84-A7-5B-E1, Payload = CNReq+d:/xx.txt)
RX(TargetMAC = 60-CF-84-A7-5B-E1, SourceMAC = abc, Payload = CNRes++)
checkTargetMAC
DLDataInd(Payload = CNRes++)
PDataInd(Payload = CNRes++)
decodieren
CNCnf(Result = +)
Filename(FName = d:/xx.txt)
wait1msec
wait8Bit
Transmit(Line = hallo erst 'mal)
DCNReq(Lines = 3)
wait4DCNCnf
PDataReq(Payload = Filename+d:/xx.txt)
PDataReq(Payload = Transmit+hallo erst 'mal)
PDataReq(Payload = DCNReq+1)
DLDataReq(Payload = Filename+d:/xx.txt)
DLDataReq(Payload = Transmit+hallo erst 'mal)
DLDataReq(Payload = DCNReq+1)
TX(TargetMAC = abc, SourceMAC = 60-CF-84-A7-5B-E1, Payload = Filename+d:/xx.txt)
TX(TargetMAC = abc, SourceMAC = 60-CF-84-A7-5B-E1, Payload = Transmit+hallo erst 'mal)
TX(TargetMAC = abc, SourceMAC = 60-CF-84-A7-5B-E1, Payload = DCNReq+1)
tm(3000)